

Lukáš Randuška

Data Science · Space Weather Research

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PROFILE

Undergraduate student in Data Analysis and Artificial Intelligence with a research focus on the intersection of data science and space physics. Currently a research intern at the Institute of Experimental Physics of the Slovak Academy of Sciences, working on space weather data analysis and the prediction of geomagnetic storms and coronal mass ejections (CMEs). Co-author of two peer-reviewed publications in *Earth and Space Science*. Strong foundation in mathematics, scientific computing, and electronics, with the goal of building professional experience in space technology and physics-driven data research.

EDUCATION

Pavol Jozef Šafárik University in Košice

2024 – present

Undergraduate studies, Data Analysis and Artificial Intelligence — Košice, Slovakia

Secondary Vocational School of Electrical Engineering, Košice

2020 – 2024

Department of Information and Network Technologies — Košice, Slovakia

- Completed with the *maturita* (secondary school-leaving examination).
- Final-year project and SOČ (Secondary School Professional Activity) advanced to the national finals in 2023/2024.

RESEARCH EXPERIENCE

Research Intern

Jul 2023 – present

Institute of Experimental Physics, Slovak Academy of Sciences — Košice, Slovakia

- Perform data analysis of space weather and develop prediction models for geomagnetic storms.
- Estimate the propagation speed and onset times of coronal mass ejections (CMEs).
- Study solar wind behaviour and the topology and structure of CMEs.
- Apply Python-based scientific computing (NumPy, pandas, SciPy) to large heliophysics datasets.
- Contributed to research resulting in two co-authored peer-reviewed publications.

PUBLICATIONS

Co-author of the following peer-reviewed articles in *Earth and Space Science* (AGU):

- Mackovjak, Š., Kostárová, S., Chum, J., et al. (2026). "Dataset of Quiet Space Weather Periods for Ionospheric Studies." *Earth and Space Science*, 13, e2025EA004537. DOI: [10.1029/2025EA004537](https://doi.org/10.1029/2025EA004537)
- Kisvárdai, I., et al. (2025). "Analysis of 42 Years of Cosmic Ray Measurements by the Neutron Monitor at Lomnický štít Observatory." *Earth and Space Science*, 12, e2024EA003656. DOI: [10.1029/2024EA003656](https://doi.org/10.1029/2024EA003656)

PROJECTS

MWP — Music Without Pain

2023/2024

Final-year project & SOČ competition entry

- Software project developed as final-year and SOČ work; advanced to the national finals of the SOČ competition (2023/2024).
- Source code: github.com/peto59/MWP

TECHNICAL SKILLS

Programming: Python (proficient — NumPy, pandas, SciPy); Java (intermediate — two semesters of algorithms, data structures, and dynamic programming).

Focus areas: scientific computing, data analysis, space weather / heliophysics data, electronics.

LANGUAGES

Slovak (native) · **English** (B1) · **German** (Deutsches Sprachdiplom — DSD Pro I)

AWARDS & ACTIVITIES

Hack Košice — Hackathon

2021, 2022, 2023, 2026

- 2021 — 2nd place overall. · 2022 — 1st Sponsor Prize (Telekom).

A CME detection-to-impact pipeline for geomagnetic storm prediction

Lukáš Randuška¹, Šimon Mackovjak¹

¹ *Institute of Experimental Physics, Slovak Academy of Sciences, Košice, Slovakia*

Detection and prediction of Interplanetary Coronal Mass Ejections (ICMEs), a main driver of geomagnetic storms, is a well-known and difficult problem. Storm predictions triggered by in-situ solar-wind measurements at L1 (e.g. DSCOVR, ACE) typically offer only about an hour of lead time, while coronagraph-based methods provide longer lead but with large arrival-time uncertainty. The most informative signal for a neural-network forecast would be the velocity and density profile from onset through transit. For this reason, we are building a full detection-and-prediction pipeline. Real-time CME detection from LASCO coronagraph images, velocity and onset-time approximation, transit time prediction, and finally the generation of a synthetic transit profile to feed a Dst/SYM-H prediction model. CME detection is done with CACTus-inspired approach, fitting a straight line to a ridge in a J-map, more precisely bright features in the J-map are thresholded and morphologically grouped into clusters, to which a RANSAC regression line is fitted; this stage reaches a velocity mean absolute error of 160 km/s, consistent with the standard CACTus method and validating it as the pipeline's front end; the parameters of this line then approximate velocity and onset time. We next use a physics-informed neural network (PINN) to predict transit time, which is in turn used to reconstruct the velocity time-series during transit, from onset to arrival. This gives the prediction model a much earlier warning, and information about the shape of the incoming impact, than current operational technologies. This approach lets us predict the magnitude of a geomagnetic storm not just a few hours ahead, but 10 or more hours — potentially more than a day — and directly addresses the ESWW2026 theme of combining data-driven and physics-based, cross-scale forecasting.

Tracking Code = [ESWW26-ABS-5566-629-158660-20260626112610]



Faculty of Science
Šrobárova 2, 040 01 Košice

ENROLLMENT CERTIFICATE

This is to certify that Lukáš Randuška, born in Košice, on 2005-01-04, is, in the 2025/2026, pursuant to Act No. 131/2002 Coll. on universities and on amendments of certain laws, as amended, a regular student enrolled in the 2nd year of study, study program Data Science and Artificial Intelligence, Full-Time form, Bachelor level of study, 1st level of tertiary study with the standard length of the study 3 years.

Enrollment date in the academic year 2025/2026: 2025-09-01.

This certificate is issued at the student's request and serves for official purposes.

Košice, 2026-06-30

Univerzita Pavla Jozefa Šafárika v Košiciach
Prírodovedecká fakulta, Šrobárova 2
040 01 Košice

Stamp and signature (Study Department)

Lukáš Randuška

Institute of Experimental Physics, Slovak Academy of Sciences

Košice, Slovakia

lukasran28@gmail.com

26 June 2026

Dear ESWW2026 Selection Committee,

I am a Bachelor student of Data Science and Artificial Intelligence at Pavol Jozef Šafárik University in Košice. I am currently working part-time at the Institute of Experimental Physics, Slovak Academy of Sciences, in Košice.

For the past three years I have been researching coronal mass ejections (CMEs), the solar wind, and their interactions with one another and with the Earth — how they affect our planet and how their behaviour can be predicted. I am currently developing a CME detection-to-impact pipeline. It begins with a CACTus-inspired CME detection algorithm applied to LASCO C2 coronagraph images, reaching effectiveness on par with the original CACTus algorithm. My approach then ties this detection directly to a network that predicts the transit time of the CME from parameters observable at onset. This network is a physics-informed neural network, pre-trained on synthetic events generated as solutions of the extended drag-based model (EDBM). The detected velocity, onset time, and predicted transit time are then used to reconstruct a 1D time-series of the CME's transit velocity up to its arrival at Earth, which we expect will yield strong results in predicting the Dst index tens of hours into the future.

I am coming to ESWW2026 to present my work and to gain experience in communicating my research. The conference would let me present my results to the international space-weather community, receive constructive criticism from experts in the field, and find potential collaborators working on CME detection and forecasting. As an early-career researcher standing at the start of my path, I am also keen to see the current frontier of machine-learning research in space weather, which would directly help guide and inspire the direction of my own work and my plans for graduate study. This would be my first international conference, and attending it would be a formative step in my training.

Thank you for considering my application. I would be very grateful for the opportunity to take part in ESWW2026 and to contribute to the space-weather community.

Sincerely,

Lukáš Randuška



INSTITUTE OF MATHEMATICS

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Recommendation Letter

To Whom It May Concern

From: RNDr. Jaroslav Šupina, PhD., Delegate of Slovakia to S2P
Institute of Mathematics, Faculty of Science, P.J. Šafárik University in Košice

Lukáš Randuška is an undergraduate student at the Institute of Mathematics, Faculty of Science, P.J. Šafárik University in Košice. His study program is Data Science and Artificial Intelligence. Although I know him personally as a passionate fan of space research from local space community meetups, I do have a professional experience with him as well. He took a mandatory course in mathematical analysis I am teaching and delivered extraordinary performance that could be hardly comparable to most of his classmates. He opted for a thesis under my supervision as well.

With the arrival of New Space we need more than anytime before those who are highly passionate in space sector on one hand but are capable to execute on the other hand. Despite his young age, Lukáš has years of experience in performing investigations in space weather domain while he obtained real research skills. That was imminent during our discussions on thesis selection and initial work in that frame. Here comes the second part of his competences, the one obtained with his studies. His deep understanding of neural networks and capabilities to apply the methods of artificial intelligence is a promising contribution to space weather research. His planned poster is a suitable example of that fact.

All the facts mentioned above entitle me to recommend to support this young promising talent. He currently does not have any other means to cover his participation at the European Space Weather Week taking place in Florence.

Thank you for your support.

Yours Sincerely,

Jaroslav Šupina

Assistant Professor & Delegate of Slovakia to S2P
Institute of Mathematics
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RNDr. Šimon Mackovjak, PhD.
Department of Space Physics, Institute of Experimental Physics,
Slovak Academy of Sciences, Watsonova 47, 040 01 Košice, Slovakia
mackovjak@saske.sk



June 30th, 2026

Dear Selection Committee,

I am pleased to provide a strong recommendation for Lukáš Randuška in support of his application for financial assistance to attend the European Space Weather Week 2026. I have been supervising Lukáš at the Institute of Experimental Physics of the Slovak Academy of Sciences in Košice, where he has been working as an intern alongside his Bachelor's studies in Data Science and Artificial Intelligence at Pavol Jozef Šafárik University.

Lukáš has spent the past three years developing a comprehensive CME detection-to-impact pipeline for geomagnetic storm prediction — a task that would challenge many postgraduate researchers, let alone an undergraduate. His pipeline begins with a CACTus-inspired CME detection algorithm applied to LASCO C2 coronagraph images, which he validated to perform on par with the original CACTus method, achieving a velocity mean absolute error of 160 km/s. Crucially, Lukáš went beyond detection: he coupled this front end to a physics-informed neural network (PINN), pre-trained on synthetic events generated as solutions of the extended drag-based model, to predict CME transit time from parameters observable at onset. The predicted transit time is then used to reconstruct a 1D velocity time-series of the CME's journey from Sun to Earth, providing upstream input for a Dst/SYM-H prediction model. This end-to-end approach directly addresses the central challenge of space weather forecasting: generating accurate geomagnetic storm warnings not just minutes, but potentially more than a day, before impact.

What distinguishes Lukáš is not only the technical depth of his work, but the independence and initiative with which he has pursued it. Over three years, he has built up a serious understanding of solar wind physics, machine learning, and operational forecasting — a rare combination at the undergraduate level. His contribution aligns squarely with the ESWW 2026 theme of combining data-driven and physics-based, cross-scale forecasting, and the conference represents an excellent opportunity for him to present his results to an international audience, receive constructive feedback from leading experts in the field, and identify potential collaborators. This would be his first international conference, and I believe attending it would be a formative step in his development as a researcher.

I wholeheartedly support Lukáš's application for financial support and recommend him without reservation. Should you require any further information, please do not hesitate to contact me.

Yours sincerely,

RNDr. Šimon Mackovjak, PhD.